ĐẠI HỌC QUỐC GIA TP. HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC BÁCH KHOA
KHOA KHOA HỌC VÀ KỸ THUẬT MÁY TÍNH



BÀI THÍ NGHIỆM 1

MÔN HỌC: HỆ THỐNG SỐ

Nhóm: 91

GVHD: Phan Đình Thế Duy

Sinh viên:

Họ và tên	MSSV
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Phạm Minh Đức	2310797
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2023

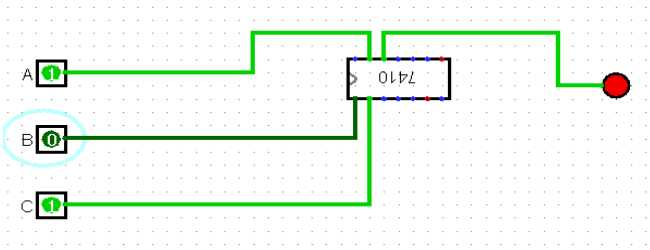
2.4.1

1. IC: 7410, 7420.

2&3. Truth tables, Logisim:

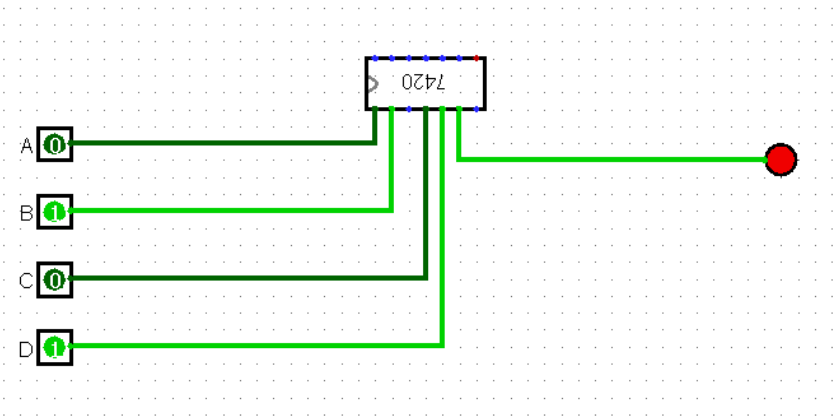
IC 7410

A	B	C	Y=(ABC)’
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



IC 7420

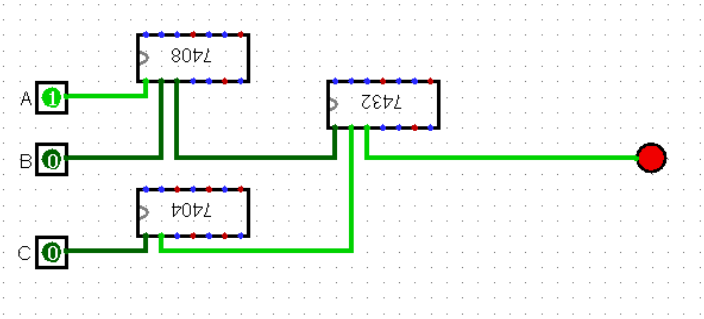
A	B	C	D	Y=(ABCD)’
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



2.4.2

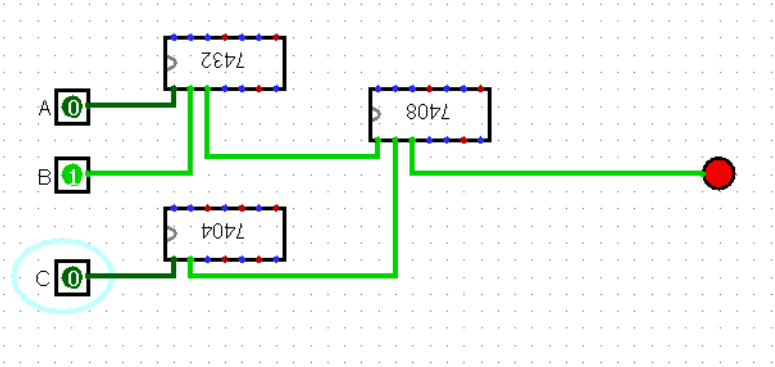
a. X= AB + C’

A	B	C	X=AB+C’
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



b. Y= (A+B)C’

A	B	C	Y=(A+B)C’
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

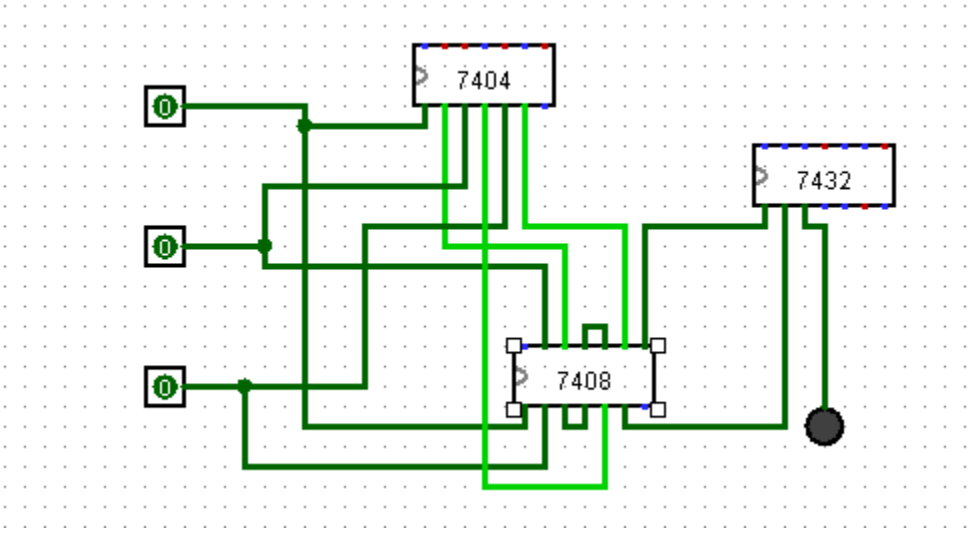


2.4.3

a. Z=(M+N)(M’+P)(N’+P’)

=N.M’.P’+M.N’.P

M	N	P	Z=N.M’.P’+M.N’.P
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0



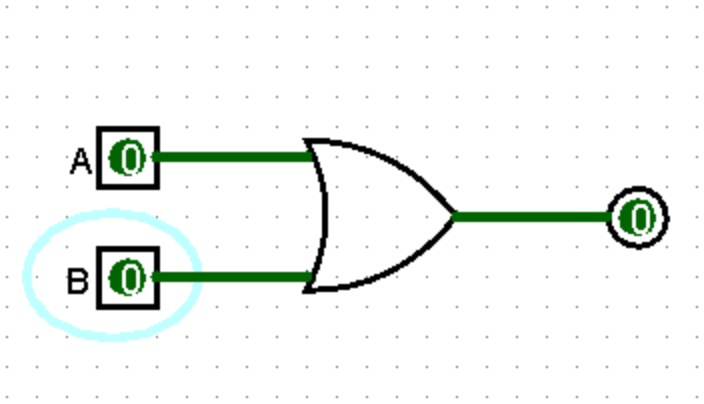
P \ MN	00	01	11	10
0	0		0	0
1	0	0	0	

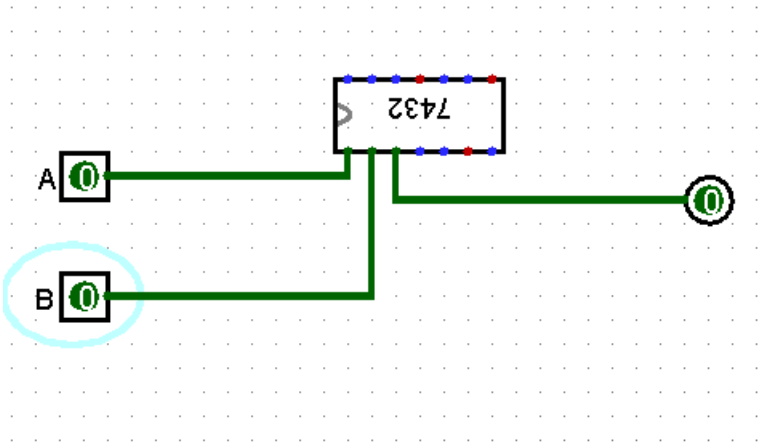
W=A’(A+B)+(B+AA)(A+B’)

=B+A

A	B	Y=B+A
0	0	0
0	1	1
1	0	1
1	1	1

A\B	0	1
0	0	1
1	1	1





Netlist

2.4.1

IC1:7410

IC 7410

SW1-->IC1:13

SW2-->IC1:1

SW3-->IC1:2

IC1:12-->LED1

IC1:7420

SW1-->IC1:1

SW2-->IC1:2

SW3-->IC1:4

SW4-->IC1:5

IC1:6-->LED1

2.4.2

a. IC1:7408;IC2:7404;IC3: 7432

SW1-->IC1:1

SW2-->IC1:2

SW3-->IC2:1

IC1:3-->IC3:1

IC2:2-->IC3:2

IC3:3-->LED1

b.IC1:7432; IC2:7404; IC3:7408

SW1-->IC1:1

SW2→IC1:2

SW3→IC2:1

IC1:3-→IC3:1

IC2:2→IC3:2

IC3:3→LED1

2.4.3

A) IC1:7404; IC2:7408; IC3:7408

SW1→IC1:1

SW1→IC2:1

SW2→IC1:3

SW2→IC2:13

SW3→IC1:5

SW3→IC2:2

IC1:2→IC2:12

IC2:11→IC2:10

IC1:6→IC2:9

IC2:3→IC2:4

IC1:4→IC2:5

IC2:6→IC3:1

IC2:8→IC3:2

IC3:3→LED1

B) IC1:7432

SW1-->IC1:1

SW2-->IC1:2

IC1:3-->LED1